DELHI PUBLIC SCHOOL NEWTOWN
SESSION 2025–26
PRE BOARD EXAMINATION

CLASS: X **FULL MARKS: 80**
SUBJECT: MATHEMATICS [SET A] **TIME: 3 HOURS**

Answers to this Paper must be written on a paper provided separately.

You will not be allowed to write during the first 15minutes.

This time is to be spent in reading the question paper.

The time given at the head of this Paper is the time allowed for writing the answers.

Attempt all questions from Section A and any four questions from Section B.

The intended marks for questions or parts of questions are given in brackets [ ].

This paper consists of eight printed pages.

SECTION A

(*Attempt **all** questions from this section)*

Question 1

Choose the correct answers to the questions from the given options. (Do not copy the questions, write the correct answer only.) **[15]**

(i) A retailer purchases an item at a marked price of ₹ 15000 from a wholesaler and sells it to a customer at 10% profit. The sales are intra-state and the rate of GST is 18%. The amount of GST paid by the retailer is:

(a) ₹ 150 **(b) ₹ 165** **(c) ₹ 210** **(d) ₹ 270**

(ii) If x = 1 is a common root of the equation $ax^2 + ax + 2 = 0$ and $x^2 + x + b = 0$ then the value of *ab* is:

(a) 1 **(b) 2** **(c) 4** **(d) 3**

(iii) If A = $[a_{ij}]$ is a 2 × 3 matrix such that $a_{ij} = \frac{(-i+2j)^2}{5}$, then a_{23} is:

(a) $\frac{1}{5}$ **(b) $\frac{2}{5}$** **(c) $\frac{9}{5}$** **(d) $\frac{16}{5}$**

(iv) If in two triangles ΔDEF and Δ PQR, $\angle D = \angle Q$ and $\angle R = \angle E$, then which of the following is *not* true?

(a) $\frac{DE}{QR} = \frac{DF}{PQ}$ **(b) $\frac{EF}{PR} = \frac{DF}{PQ}$** **(c) $\frac{DE}{PQ} = \frac{EF}{RP}$** **(d) $\frac{EF}{RP} = \frac{DE}{QR}$**

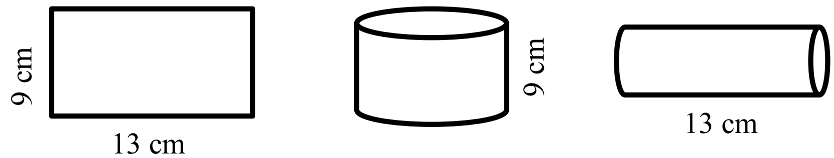
(v) A rectangular sheet of paper of size 13 cm × 9 cm is first rotated about the side 13 cm and then about the side 9 cm to form a cylinder, as shown in the diagram. The ratio of their curved surface areas is:

(a) 1 : 1

(b) 9 : 13

(c) 13 : 9

(d) $\frac{13\pi}{9} : \frac{9\pi}{13}$



(vi) For the following distribution

Marks	Below 10	Below 20	Below 30	Below 40	Below 50	Below 60
Cumulative Frequency	3	12	27	57	75	80

The modal class is:

(a) 10 - 20 (b) 20 - 30 (c) 30 - 40 (d) 50 - 60

(vii) If the mid-point of the line segment joining the points (a, 4) and (2, 2b) is (2, 6), then the value of (a + b) is given by:

(a) 6 (b) 7 (c) 8 (d) 16

(viii) Assertion (A) : cosecθ + cot θ = p and cosecθ - cot θ = q then, pq = 1

Reason (R) : $cosec^2\theta - cot^2\theta = 1$

(a) Both A and R are true, and R is the correct reason for A.

(b) Both A and R are true, and R is incorrect reason for A.

(c) A is true, R is false. (d) A is false, R is true.

(ix) Assertion (A) : $x^3 + 2x^2 - x - 2$ is a polynomial of degree 3.

Reason (R) : x + 2 is a factor of the polynomial.

(a) Both A and R are true, and R is the correct reason for A.

(b) Both A and R are true, and R is incorrect reason for A.

(c) A is true, R is false. (d) A is false, R is true.

(x) A coin is tossed and a die is rolled. The probability that the coin shows a head and the die shows 3 is:

(a) $\frac{1}{6}$ (b) $\frac{1}{9}$ (c) $\frac{1}{11}$ (d) $\frac{1}{12}$

(xi) A person deposited ₹ 700 per month for 18 months. The qualifying sum of money for the calculation of interest is:

(a) ₹ 6300 (b) ₹ 12,600 (c) ₹ 67550 (d) ₹ 1,19,700

(xii) Which is a better investment?

(I)16% ₹ 100 shares at ₹ 80 (II) 20% ₹ 100 shares at ₹ 120

(a) only I. (b) only II.

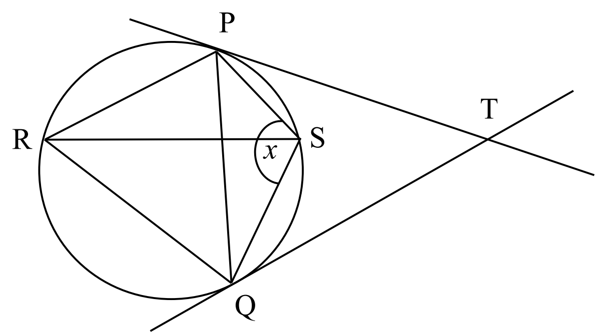
(c) both I and II have equal values. (d) can't be determined.

(xiii) If the replacement set is the set of integers between -10 and 10, then the solution set of the inequation ; $-2 < \frac{1}{2} - \frac{2x}{3} < 1\frac{5}{6}$ is:

(a) { -1, 0, 1, 2} (b) { -2,-1, 0, 1, 2, 3} (c) { -1, 0, 1, 2, 3} (d) { -2,-1, 0, 1, 2}

(xiv) In the given figure, PT and QT are tangents to a circle such that ∠TPS = 45° and ∠TQS = 30°. Then the value of x is:

(a) 30°
(b) 45°
(c) 75°
(d) 105°



(xv) Statement 1 : If a, b, c, d are in proportion then $\frac{a}{b} = \frac{c}{d} \Rightarrow$ ad = bc

Statement 2 : If a, b, c are in continued proportion, then b = $\sqrt{ac}$

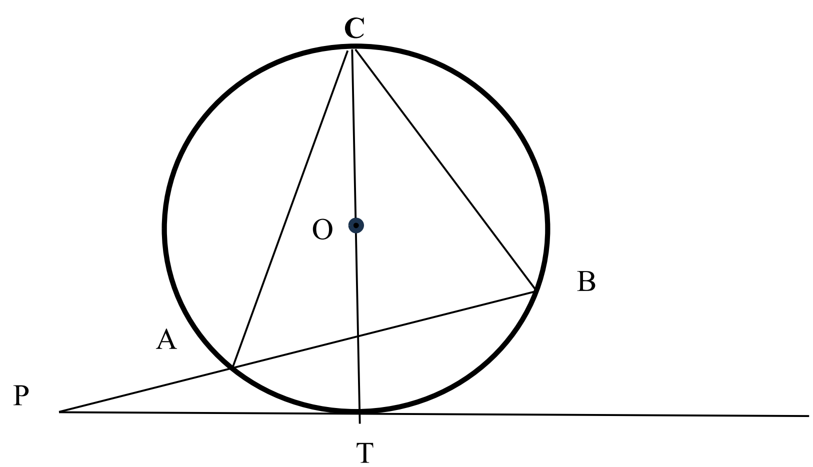
Which of the following is valid?

(a) Both statements are true.
(b) Both statements are false.
(c) Statement 1 is true and Statement 2 is false.
(d) Statement 1 is false and Statement 2 is true.

Question 2

(i) In the given figure, ∠ABC = 70° and ∠ ACB = 50°. Given, O is the centre of the circle and PT is the tangent to the circle. Then calculate the following angles:

(a) ∠CBT
(b) ∠BAT
(c) ∠PBT
(d) ∠APT



[4]

(ii) Use remainder and factor theorem, show that (2x +3) is a factor of the polynomial $2x^2 + 11x + 12$. Hence, factorise it completely. What must be multiplied to the given polynomial so that $x^2 + 3x - 4$ is a factor of the resulting polynomial? Also write the resulting polynomial. **[4]**

(iii) Given the equations of two straight lines, L_1 and L_2 are $x - y = 1$ and $x + y = 5$ respectively. If L_1 and L_2 intersects at point Q (3, 2). Find :

(a) the equation of line L_3 which is parallel to L_1 and has y-intercept 3.

(b) the value of k, if the line L_3 meets the line L_2 at a point P (k,4).

(c) the coordinate of R and the ratio PQ: QR, if line L_2 meets X-axis at point R. **[4]**

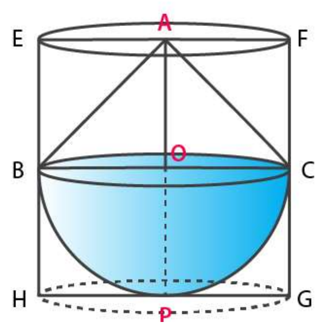
Question 3

(i) (a) Write the nth term(T_n) of an Arithmetic Progression consisting of all whole numbers which are divisible by 3 and 7.

(b) How many of these are two- digit numbers? Write them.

(c) Write the sum of first 10 terms of this A.P. **[4]**

(ii) A solid toy is in the form of a hemisphere surmounted by a right circular cone. The height of the cone is 2 cm, and the diameter of the base is 4 cm. Determine the volume of the toy. If a right circular cylinder circumscribes the toy, find the difference between the volumes of the cylinder and the toy.



[4]

(iii) (Use ruler and compasses only for this question.)

(a) Construct the locus of a moving point which moves such that it keeps a fixed distance of 4.5 cm from a fixed point O.

(b) Draw a line segment AB of 6 cm where A and B are two points on the locus (a).

(c) Construct the locus of all points equidistant from A and B. Name the points of intersection of the loci (a) and (c) as P and Q respectively.

(d) Join PA. Find the locus of all points equidistant from AP and AB.

(e) Mark the point of intersection of the locus (a) and (d) as R. Measure and write down the length of AR. **[5]**

SECTION B

(*Attempt* ***any*** ***four*** *questions from this section)*

Question 4

(i) A man bought ₹ 200 shares of a company at 25% premium. If he received a return of 5% on his investment. Find the:
(a) market value
(b) dividend percent declared
(c) number of shares purchased, if annual dividend is ₹1000. **[3]**

(ii) The profit in rupees in a local restaurant and the number of customers who visited the restaurant are tabulated below for each week for one month.

Week number	Week 1	Week 2	Week 3	Week 4
No: of customers	1400	5600	x	3212
Profit in ₹	28000	112000	32140	y

Find:
(a) If the number of customers and profit per week is in continued proportion or not. Justify your answer.
(b) the value of x and y. **[3]**

(iii) Find:
a) $(\sin\theta + \operatorname{cosec}\theta)^2$ b) $(\cos\theta + \sec\theta)^2$
Using the above results prove the following trigonometry identity.
$(\sin\theta + \operatorname{cosec}\theta)^2 + (\cos\theta + \sec\theta)^2 = 7 + \tan^2\theta + \cot^2\theta$ **[4]**

Question 5

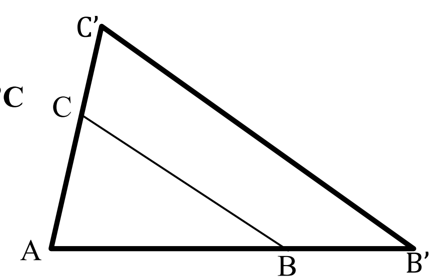
(i) A man has a recurring deposit account in a bank and he deposits ₹500 per month for a period of 4 years. If he gets ₹ 28410 on maturity, find the rate of interest. **[3]**

(ii) Using step-deviation method, find the mean of the following frequency distribution.

Class	11 - 20	21 - 30	31 - 40	41 - 50	51 - 60	61 - 70	71 - 80
Frequency	2	6	10	12	9	7	4

[3]

(iii) While preparing a Power Point presentation, Δ ABC is enlarged along the side BC to Δ AB'C', as shown in the diagram, such that BC : B'C' is 3:5 and BC||B'C'. Find:
(a) AB : BB'
(b) Length AB, if BB' = 4cm
(c) Is Δ ABC ~ Δ AB'C'? Justify your answer.
(d) Area of Δ ABC : Area of Quadrilateral BB'C'C



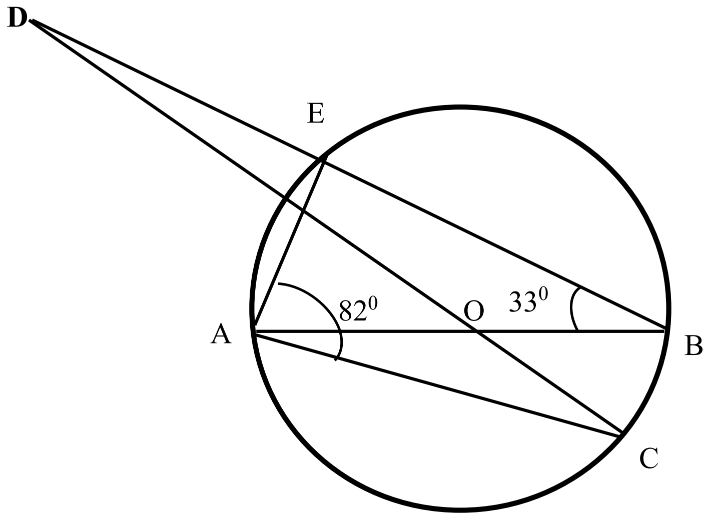
[4]

Question 6

(i) **If the midpoint of the line segment joining the points A(3, 4) and B(k, 6) is P(x, y) and the equation of a line through P is x + y - 10 = 0, find the value of k.** **[3]**

(ii) **AB and CD intersect at the centre O of the circle given in the above diagram. If $\angle EBA = 33^0$ and $\angle EAC = 82^0$, find:**

(a) $\angle BAE$
(b) $\angle BOC$
(c) $\angle ODB$



[3]

(iii) **Refer to the given bill.**
A customer paid ₹ 2000 (rounded off to the nearest ₹ 10) to clear the bill.
Note: 5% discount is applicable on an article if 10 or more articles are purchased.

BILL			
Article	**M.P.(₹)**	**Quantity**	**G.S.T.**
A	**190**	**06**	**12%**
B	**50**	**12**	**18%**

Check whether the total amount paid by the customer is correct or not. Justify your answer with necessary working. **[4]**

Question 7

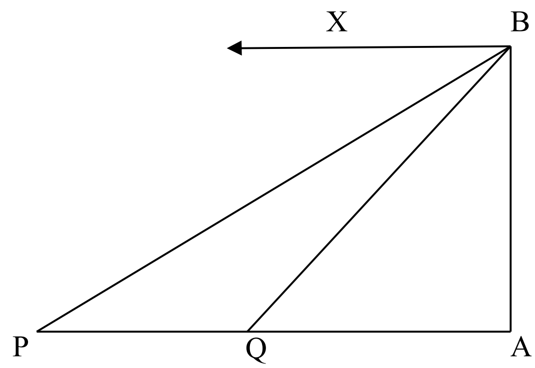
(i) **A health insurance agent found the following data for distribution of ages of 100 policy holders:**

Age(in years)	**20-25**	**25-30**	**30-35**	**35-40**	**40-45**	**45-50**	**50-55**	**55-60**
No: of Policy Holders	**2**	**4**	**12**	**20**	**28**	**22**	**8**	**4**

On a graph sheet draw an ogive using the given data. Take 2cm = 5 years along one axis and 2cm = 10 policy holders along the other axis. Use your graph to find:

(a) The median and upper quartile.
(b) Number of policy holders whose age is above 52 years. **[5]**

(ii) A lighthouse, AB, stands tall on a cliff by the sea, watching over ships that pass by. One day a ship is seen approaching the shore and from the top of the lighthouse, the angles of depression of the ship are observed to be 30^0 and 45^0 as it moves from point P to point Q. The height of the lighthouse is 50 metres.



Based on the information given above, answer the following questions:

(a) Find the distance of the ship from the base of the lighthouse when it is at point Q, where the angle of depression is 45^0.

(b) Find the distance travelled by the ship between points P and Q.

(c) If the ship continues moving towards the shore and takes 10 minutes to travel from Q to A, calculate the speed of the ship in km/hr, from Q to A. **[5]**

Question 8

(i) Solve the following inequation, write down the solution set and represent it on the real number line.

$$- 3 + x \leq \frac{7x}{2} + 2 < 8 + 2x, x \in I$$ **[3]**

(ii) Fifty seeds were selected at random from each of 5 bags of seeds, and were kept under standardized conditions favourable to germination. After 20 days, the number of seeds which had germinated in each collection were counted and recorded as follows:

Bag	1	2	3	4	5
Number of seeds germinated	40	48	42	39	41

What is the probability of germination of

(a) atleast 40 seeds in a bag?

(b) 49 seeds in a bag?

(c) more than 35 seeds in a bag? **[3]**

(iii) It costs ₹ 2200 to paint the inner curved surface area of a cylindrical vessel 10m deep. If the cost of painting is ₹ 20 per m^2, find:
(a) inner curved surface area of the vessel,
(b) radius of the base,
(c) capacity of the vessel. **[4]**

Question 9

(i) The sum of the first three terms of a G.P. is $\frac{39}{10}$ and their product is 1. Find the common ratio and the terms. **[3]**

(ii) Solve for 'x' using quadratic formula: $2x^2 - 4 = 5x$. Give your answer correct to three significant figures. (*Use Mathematical tables for this question)* **[3]**

(iii) The Indian Meteorological Department observes seasonal and annual rainfall every year in different sub-divisions of our country. It helps them to compare and analyze the results. The table below shows sub-divisions wise seasonal rainfall (in mm).

Rainfall(mm)	200-400	400-600	600-800	800-1000	1000-1200	1200-1400
No: of Sub-divisions	3	4	7	5	4	3

Based on the information given above, answer the following questions:
(a) Write the Modal Class.
(b) Draw a histogram representing the above distribution and estimate the mode from the graph. **[4]**

Question 10

(i) Car A travels x km for every litre of petrol, while car B travels (x + 5) km for every litre of petrol.
(a)Write down the number of litres of petrol used by car A and car B in covering a distance of 400 km.
(b)If car A uses 4 litres of petrol more than car B in covering 400 km, write down the equation in x and solve it to determine the number of litres of petrol used by car B for the journey. **[3]**

(ii) If $\begin{bmatrix} 4 & -2 \\ 1 & 4 \end{bmatrix} \begin{bmatrix} x \\ 1 \end{bmatrix} + 2 \begin{bmatrix} 0 \\ -3 \end{bmatrix} = \frac{1}{3} \begin{bmatrix} 6 \\ y \end{bmatrix}$, find the values of x and y. **[3]**

(iii) Use a graph paper for this question taking 2cm = 1 unit along both axes.
(a) Plot A(1, 3), B(1, 2) and C(3, 0). Reflect A and B on the x-axis and name their images as E and D respectively. Write down their coordinates.
(b) Reflect A and B through the origin and name their images as F and G respectively.
(c) Reflect A, B, C on the y-axis and name their images as J, I and H respectively.
(d) Join all the points A, B, C, D, E, F, G, H, I and J in order and name the closed figure so formed. **[4]**